RAGHAV MANSHARAMANI

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EDUCATION

University of Illinois at Urbana-Champaign Expected May 2027
Bachelor of Science in Aerospace Engineering — Minor in Mathematics Cumulative GPA: 3.92/4.0
Dean's List: Fall 2024, Spring 2025, Fall 2025
Relevant Coursework: Compressible Flow, Thermodynamics, Statistics & Probability I & II, Numerical Methods

EXPERIENCES

SpaceX May 2026 – Aug 2026
Reliability Engineering Intern - Starlink Gateway *Bastrop, TX*

- Owned lifetime qualification of the Gateway three-phase BLDC motor via accelerated life testing (hot soak, hot humid, thermal cycling); set test durations from Arrhenius and Norris-Landzberg models and chamber capability.
- Screened motors with back-EMF, Hipot, and IR testing at each stress interval to quantify degradation; qualified the motor to a 5-year equivalent life with no observed failure modes.
- Automated motor controller commands in Python for repeatable dynamometer torque runs and back-EMF measurements, cutting manual runtime.
- Discovered a non-functional over-temperature protection circuit during battery charger HALT; presented findings at CDR to the Gateway team, driving a supplier-side design fix.

NASA Johnson Space Center Jan 2026 – May 2026
Engineering Intern – Applied Aeroscience and CFD Branch (EG3) *Houston, TX*

- Automated structured overset grid generation for HyperSTEP's newest movable-flap configuration via TCL scripting with Chimera Grid Tools (CGT); built as a reusable pipeline for future flap configurations.
- Ran 300+ high-fidelity OVERFLOW CFD cases on HPC clusters across sub-, trans-, and supersonic regimes to build a more accurate force/moment database for GNC's Monte Carlo reentry trajectory simulations.
- Developed Python scripts to extract, reformat, and compare force/moment coefficients across CFD solvers and wind tunnel data using Dynaplot for independent verification and validation (IV&V) of the database.
- Identified and corrected a significant subsonic/transonic error in GNC's existing Cart3D-based database by validating against higher-fidelity OVERFLOW data; presented findings to HyperSTEP project leadership.

Luminary Cloud CFD May 2024 – July 2024
Physics Team Intern *San Mateo, CA*

- Meshed and ran CFD simulations of a generic automotive model for the Automotive CFD Prediction Workshop (AutoCFD), testing solver configurations (e.g. DDES vs. DES turbulence models, k-ω vs. k-ϵ).
- Analyzed CFD results in Python to produce V&V documentation presented to prospective customers and informed the team's future meshing and geometry setups for AutoCFD.
- Configured and tested Luminary's new mesh adaptation feature on 10+ geometries; presented findings to the engineering team to be used as customer-facing examples.

ACTIVITIES

Illini Electric Motorsports – Formula SAE Sep 2023 – May 2025
Project Lead – Aerodynamics Team *Champaign, IL*

- Spearheaded design of the front wing mainplane via parameterized CAD in PTC Creo, conditioning airflow to undertray and rear wing; increasing front wing downforce by 15% and decreasing total car drag by 3%.
- Co-led rear wing gurney flap design using parameterized CAD models; ran CFD simulations across 15+ gurney flap and mainplane designs to identify the mainplane angle of attack and gurney flap size giving max downforce for least drag.
- Contributed 80+ hours in prepreg carbon-fiber layups alongside 15+ members to manufacture ∼20 aerodynamic elements; post-processed aerodynamic components (trimming, sanding, finishing) for final assembly.

TECHNICAL SKILLS

Star-CCM+, OVERFLOW, NX, Python, Pointwise, Linux, ANSYS Mechanical/ACP, Composite Manufacturing